Qena STEM school

Group:- 23304

ITO+ *plus*

Intelligent Traffic Observer

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ABSTRACT

Egypt faces a myriad of problems, among which is the escalating issue of urban congestion, a national challenge driven by rapid population growth, inefficient traffic control, and the increasing density of vehicles on the roads, these factors increase trip time, fuel consumption & pollution emission. Which has a major contribution in critical drawbacks worldwide such as the economic aspects, spreading air pollution, and declining public health. this project approaches these challenges through a novel IoT & AI-based intelligent traffic system, starting with the road system, the road represents an intersection, simulating real-life situations. The road is applied with sensors and camera for processing cars and extracting data like the cars speed, distance covered, accident detections, and best routing options. These data are extracted and sent to the backend server for analysis, enabling dynamic traffic light control. Processed data are sent to the vehicles via a user-friendly mobile dashboard applying the V2V & V2I communication. The system proved its efficiency in improving the traffic flow, trip time, and pollution reduction by reducing congestion, detecting the fastest routing option for the vehicles. it achieved the required design requirement by having a 1.23s response time, maintaining a regulated cars density level. Proved system accuracy confirms its reliability and effectiveness for intelligent transportation applications.

INTRODUCTION

Egypt is currently facing severe urban and environmental challenges due to increasing traffic congestion, rising vehicle density, and continuous urban expansion. These problems have been accelerated significantly by the rapid population growth rate of 1.5 million people annually, the high urbanization rate exceeding 43%, and the concentration of more than 22 million residents in Greater Cairo alone, all contributing to extensive pressure on road networks as shown in **Fig (1)**.[6] Extended congestion has been shown increase fuel consumption by 20-40%, elevate CO_2 and NO_x emissions by up to 30%, raise accident rates, and reduce overall road efficiency **(United Nations, 2021)**. A major contributor to these issues is the absence of adaptive traffic-management systems capable of responding to real-time changes in vehicle flow, resulting in increased delays, reduced safety, and limited emergency-vehicle mobility, particularly in densely populated areas such as Cairo, Giza, and Alexandria as shown in **Fig (2)**.[4]

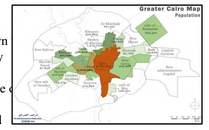


Fig. 1 -Growing population of Greater Cairo (Source: marsadomran.info)

Consequently, several intelligent transportation systems have been implemented worldwide in attempts to mitigate congestion. Systems such as the Cairo ITS[2], Singapore ITS, Seoul TOPIS, and the New Administrative Capital ITS have demonstrated strengths including high reliability, real-time surveillance, adaptive signal control, and effective violation detection, proving their durability and large-scale efficiency. However, their limitations have also been documented. These systems exhibit extremely high installation and operational costs averaging millions of dollars per zone, limited coverage outside central regions, dependence on dense infrastructure, lack of V2V communication, absence of real-time accident prediction, and in many cases no measurable reduction in vehicle volume, meaning congestion persists even with monitoring **(LTA Singapore, 2021)**. These strengths and weaknesses collectively revealed the need for a low-cost, scalable, and fully automated system aligned with the Capstone Challenge requirements of communication, sensing, information, and informatics.

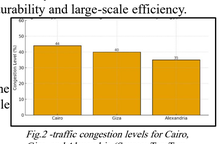


Fig.2 -traffic congestion levels for Cairo, Giza, and Alexandria (Source: TomTom Traffic Index)

This prototype tends to solve these limitations by integrating a fully automated intelligent-road system supported by machine learning, IoT communication, and real-time decision making. The system requires low operational cost, produces no harmful byproducts, and is equipped with continuous monitoring and adaptive control. The road-management process comprises several stages. The first stage is real-time sensing, where a camera module, IR speed-detection gates, and MQ-135 air-quality sensors continuously measure vehicle count, speed, pedestrian flow, and CO_2 concentration to determine congestion level. The second stage depends on ML algorithms that perform vehicle recognition, emergency-vehicle detection, and accident prediction. The third stage focuses on intelligent control, where signal timings and lane permissions are adjusted automatically. Another essential stage relies on V2V and V2I communication, allowing smart vehicles to harmonize with the road system for enhanced safety and routing.

This prototype is characterized by an integrated IoT-based feedback design. Instead of fixed timer intersections, real-time road conditions govern each decision. At the end of each sensing cycle, data such as vehicle speed, congestion density, and pollution level are collected and analyzed. Based on these readings, an automated feedback loop determines whether to extend lights, shorten them, activate the emergency green corridor, or maintain the current flow. Continuous cycles are performed to increase accuracy and minimize environmental impact. Furthermore, the prototype consumes minimal energy, operates under various conditions, and can be expanded easily to represent more complex road networks.

The project's environmental, technological, and economic impact is significant. By reducing congestion, trip time, idling, and fuel waste, the prototype directly contributes to lowering air-pollution levels associated with crowded roads. Additionally, through the use of low-cost sensors, real-time communication, and AI-integrated processing, the system supports national efforts to improve technological infrastructure and addresses several of Egypt's Grand Challenges, including congestion reduction, pollution reduction, population-pressure mitigation, and public-health improvement. With strong adaptability, minimal energy consumption, and compliance with design requirements, this intelligent road prototype represents a durable, replicable, and environmentally friendly solution capable of advancing Egypt's transportation future

MATERIALS

Table-1

Picture	Item	Quantity	Usage	Source
	Plywood (150x150 cm)	2 Pieces	Manufacturing the car body using CNC laser cutting	FabLab
	IR Sensor	6 Pieces	Detect car speed, and line tracking for cars	Makers Electronics
	L9110H Motor driver Module	2 Pieces	Used to drive and control the motors	
	HC-SR04 Ultrasonic Sensor	2 Pieces	Obstacle avoidance, and communication between cars.	
	ESP32 Microcontroller	2 Pieces	Controlling car components and communication between cars and the system	
	Arduino Uno Microcontroller	1 Piece	Used to control the centralized system of the road	
	DC-Motor	8 Pieces	Used as wheels for the cars	
	Servo Motor	2 Pieces	Used to open and close the green corridor	
	LEDs	12 Pieces	Used for the traffic lights	Jumpers Electronics
	Camera	1 Piece	Used as a camera for the traffic system	Used

METHODS

While constructing the prototype, safety protocols were strictly followed. Gloves, masks, and coats were worn. The work was conducted in a safe and suitable environment under the supervision of lab teachers.

-Firstly, the process started by collecting the required materials that are necessary for the construction process, and then the prototype construction began. A 150cm x 150cm wooden plate was used as the road for the system. Holes were drilled to allow safe wire routing and internal connections. Then it was painted black with white stripes in the middle for the line following cars and enhance camera visibility.

-A central processing unit was installed and wired to control the traffic lights, process sensor inputs, and enable V2I

-Secondly, each vehicle was designed as a scaled smart car using ESP32 for V2X (V2V & V2I) communication by using data transfer via the Wi-Fi module. Motors were connected through an H-bridge and applied with an IR line-following algorithm for accurate path tracking. **Fig. (3)** represents the Full circuit for the line tracking car, As shown in **Fig. (4)**, which illustrates the 3D model of the designed smart vehicle, and **Fig. (5)**, which presents the actual constructed prototype.

-Thirdly, A camera was fixed securely above the prototype to obtain a clear top-down view of the entire road. To ensure accurate measurement, reference points were applied on the road so that the camera could scale pixel distances into real-life measurements. Using these references, the system detected the position, velocity, and acceleration of all moving vehicles with high precision.

-All camera parameters were calibrated to reduce errors caused by lighting or perspective distortions.

-All road-embedded sensors (IR speed sensors, traffic light modules, communication units) were connected to the central controller, enabling the V2I Communication.

-Data from the smart vehicles and the road system were continuously collected, synchronized, and transmitted to a backend server for storage and analysis. This allowed the system to record congestion levels, vehicle behavior, sensor readings, and camera-based predictions.

-A mobile application was created to act as a monitoring dashboard. The app displays all collected data All sensors and cameras were fully calibrated to ensure accurate measurements, stable readings, and reliable system operation throughout the experiment.

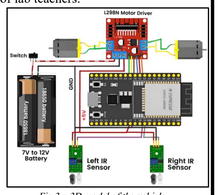


Fig.3 – 3D model of the vehicle

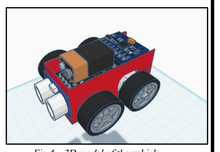


Fig.4 – 3D model of the vehicle

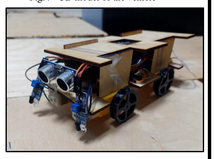


Fig.5 – actual constructed car prototype

TEST PLAN

Trip Time Improvement: Measure trip time and waiting time at intersection before and after the system implementation

Best Path Detection: Detecting the shortest and fastest path using trained ML models and algorithms, combined with the expected congestion for trip time reduction

System Accuracy: comparing different aspects of the system to measure their accuracy depending on real data. (e.g., Speed & distance detection, accident prediction, etc…)

Gas Emission Reduction: decreasing the gas emissions via the vehicles by reducing the trip time and waiting time at intersections for pollution reduction

Emergency Vehicle Detection: training an ML model to accurately detect cars and differentiate between normal vehicles and emergency vehicles.

System Response Time: The measure of time between the action occurrence and its reaction to ensure low latency and a ≤ 4-second response time for a reliable system

Car Density Handling: testing various congested cases and verifying stable congestion detection and correct system decisions for congestion resolving.

RESULTS

Table-2

	Trail 1	Trail 2	Trail 3	Trail 4	Trail 5	Trail 6	Trail 7
Waiting time Before	9.2 ± 0.2 sec	11.6 ± 0.2 sec	6.7 ± 0.2 sec	7.5 ± 0.2 sec	4.1 ± 0.2 sec	6.6 ± 0.2 sec	7.0 ± 0.2 sec
Moving Time Before	9.0 ± 0.2 sec	9.8 ± 0.2 sec	12.7 ± 0.2 sec	9.2 ± 0.2 sec	8.2 ± 0.2 sec	8.9 ± 0.2 sec	7.7 ± 0.2 sec
Trip Time Before	18.2 ± 0.2 sec	21.4 ± 0.2 sec	19.4 ± 0.2 sec	16.7 ± 0.2 sec	12.3 ± 0.2 sec	15.5 ± 0.2 sec	14.7 ± 0.2 sec
CO2 Before	21.1 ± 0.3 g	23.2 ± 0.3 g	25.0 ± 0.3 g	19.9 ± 0.3 g	16.2 ± 0.3 g	19.2 ± 0.3 g	17.8 ± 0.3 g
Waiting time After	1.2 ± 0.2 sec	0.4 ± 0.2 sec	0.7 ± 0.2 sec	0.0 ± 0.2 sec	1.5 ± 0.2 sec	3.1 ± 0.2 sec	0.5 ± 0.2 sec
Moving Time After	5.5 ± 0.2 sec	7.7 ± 0.2 sec	6.5 ± 0.2 sec	5.1 ± 0.2 sec	4.9 ± 0.2 sec	6.3 ± 0.2 sec	6.2 ± 0.2 sec
Trip Time After	6.3 ± 0.2 sec	8.1 ± 0.2 sec	7.2 ± 0.2 sec	5.1 ± 0.2 sec	6.4 ± 0.2 sec	9.4 ± 0.2 sec	6.7 ± 0.2 sec
CO2 After	9.7 ± 0.3 g	13.6 ± 0.3 g	12.3 ± 0.3 g	8.8 ± 0.3 g	8.7 ± 0.3 g	12.5 ± 0.3 g	10.9 ± 0.3 g

Table-3

	Distance before	Speed Before	Distance After	Speed After
Sec 1	40.2 ±2cm	3.6±2.3cm/s	44.9 ±2cm	14.2 ±2.3cm/s
Sec 2	59.8 ±2cm	1.3 ±2.3cm/s	66.9 ±2cm	3.6 ±2.3cm/s
Sec 3	93.4 ±2cm	1.5 ±2.3cm/s	97.4 ±2cm	3.0 ±2.3cm/s
Sec 4	112.6 ±2cm	0.0 ±2.3cm/s	127.3 ±2cm	3.2 ±2.3cm/s
Sec 5	112.6 ±2cm	0.0 ±2.3cm/s	154.1 ±2cm	1.2 ±2.3cm/s
Sec 6	112.6 ±2cm	0.0 ±2.3cm/s	--	--
Sec 7	112.6 ±2cm	0.0 ±2.3cm/s	--	--
Sec 8	112.6 ±2cm	0.0 ±2.3cm/s	--	--
Sec 9	112.6 ±2cm	0.0 ±2.3cm/s	--	--
Sec 10	112.6 ±2cm	0.0 ±2.3cm/s	--	--
Sec 11	116.4 ±2cm	1.2 ±2.3cm/s	--	--
Sec 12	120.1 ±2cm	2.3 ±2.3cm/s	--	--
Sec 13	122.9 ±2cm	2.1 ±2.3cm/s	--	--
Sec 14	130.3 ±2cm	2.4 ±2.3cm/s	--	--
Sec 15	142.8 ±2cm	2.9 ±2.3cm/s	--	--
Sec 16	154.1 ±2cm	1.3 ±2.3cm/s	--	--

Fig-5

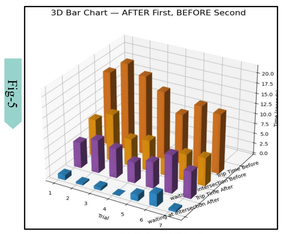
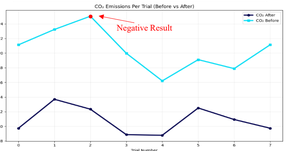


Fig-6



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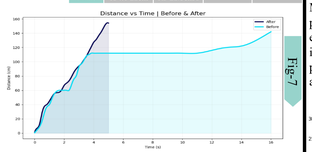
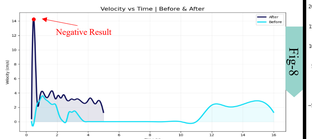


Fig-8



ANALYSIS

Due to Egypt's rapid population growth, expanding urban areas, and the increasing pressure on transportation networks, the country faces significant challenges related to urban congestion and its environmental consequences. Traffic density, industrial activities, and improper waste disposal contribute to severe pollution that affects Egypt's air, water, and soil. These conditions directly accelerate the spread of public health issues and diseases, while also placing strain on different economic and environmental sectors. Our prototype is designed to contribute to solving these national challenges and generate further positive outcomes. From the obtained results, the prototype exhaustively meets the design requirements; however, some areas still require improvement to enhance system performance and long-term impact.

Mechanism:

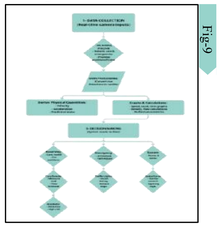
1- Vechiels detection:
data in Using a pre-trained ML model "YOLO8" for accurate cars and emergency vehicles detection. By using DeepSort, giving each detected car an ID for trajectory line used calculations Communication, for distance and velocity detection.decisions.

2- Data transmitting & receiving: Saving the Saving the JSON file & sending it to the backend server via Wi-Fi, and storing it in a database. Sending the processed data to the central processing unit and the cars via V2x, ensuring low latency for real-time control

3- Data processing:
in real time, calculating the cars collision and accident rates, processing the velocity & position of the vehicle to detect best path for the car. Also analyzing the number of cars and pedestrians at the intersection for a dynamic traffic system that enables better trip times. Processing the cars to detect emergency cars that enables the green corridor activation

4- Decision making:
The system analyzes traffic density, emergency vehicle paths, and predicted collision risk. Based on these parameters, the dynamic traffic light controller adjusts green-time durations, prioritizes emergency vehicles, and triggers alerts when abnormal speed or dangerous patterns are detected.

Fig.9



Positive results:

-system accuracy: System is mainly relying on a trained AI model to collect order and process data. However that ML algorithm is highly trained, an accuracy test was implemented to ensure the obtained results. Seven different situations tests was done on three aspects:

Fig.(10)

$$\text{Accuracy} = \frac{\text{Achieved Output}}{\text{Ideal Output}} \times 100$$

Cars detection: from the distance and speed detection test comparing AI and real tools results, AI accuracy of detection is 98.2% [1]

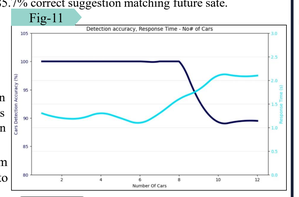
- Emergency Car Detection: System can detect emergency car existence with 100% accuracy from different motion directions.

- Speed violation: Through 4 situations with one car exceeding max speed, all cases were observed typically, 100% accuracy. However, when speed is near to max speed, system considers it reaching or exceeding it giving a wrong alarm. This problem was solved by the installed IR sensors for speed detection at the intersection, giving a refrence check point for camera periodically.

- Accident prediction: By exposing the system to accident cases, It was found to expect and resolve the accidents with a ratio of 4, 80%, cases expected and 4, 80%, resolved from 5 cases.

- Best path detection: System gives best path suggestion according to three factors, expected cars density, current cars density and shortest path. System was found to give 85.7% correct suggestion matching future sate.

Fig-11



Response time:

- Data transmitting: Response time from synthesized data transmission by camera to receiving of it by the server. Data types like (road map, cars position & speed, cars detection, speed etc …) were concluded to have an average time 1.27 seconds.

- Orders receiving: Delay time from the action of sending orders from the server (e.g. lighting traffic lights, opening green corridor gates) to when they were received was obtained as 1.29 seconds in average.

- System working performance: Delay time from sending orders action from server until they are implemented. Equipment implementation of orders was instantaneous with average from 1.3 to 1.65 seconds

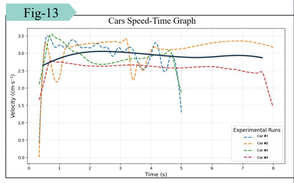
Fig.(12)

$$CO_2 = EF_{idle} \cdot t_{idle} + EF_{move} \cdot t_{move}$$

The following graph **Fig. (11)** shows values of response time and cars detection accuracy with repect to number of cars at the intersection. The graph shows that values range in a tiny range until a critical limit in congestion is reached, where detection accuracy starts slowing down while response time spikes until they stbilize at certain rates.

- Pollution Reduction:
Pollution emitted by cars was estimated considering average standard cases where an average car would consume ~0.9–1.0 L/hour at idle and emits ~125 g/km, equivalent to ~0.63 g/s during idling and ~1.73 g/s while moving. Applying the following equation in **Fig. (6)**. Our approach was to conclude with the ratio of decrement of pollution by dividing before and after cases measurements. Pollution reduction ratio was about[3]

Fig-13



-Traffic Flow: In order to deal with traffic congestion, the system has to guide Traffic flow into smoother and faster paths. The following velocity-time graph **Fig.(13)** shows traffic flow through intersection. Regression curve change in velocity, first derivative/slope **MA.3.03** ,shows the average change in velocity with respect to time. The less velocity change is, the smoother traffic flow is. The graph shows quite smooth traffic flowafter using the system, proving its efficient performance on the road.[5]

Fig.(14)

$$\text{Traffic Flow Improvement } (\%) = \left(\frac{\text{Average Velocity}_{\text{after}}}{\text{Average Velocity}_{\text{before}}}\right) \times 100$$

- Accident prediction: Using the concept of regression line studied in **MA.3.02**, system used data obtained from the camera about each car position through time and generate a regression line using parametric equations as shown in graph **Fig. (14)**. This regression line is represented in a time-extended graph of position data. Predicted accidents are the points at same time where position of two cars is the same. A sample of accident prediction cases from the tests is shown in the **fig. (14).**

- Trip time (waiting at intersections): In term of the improvement of traffic motion, the had to deal with trip time and waiting at intersections in turn. the following graph **Fig. (15)** shows trip data for each trial showing a noticeable decrease in both trip time and waiting time

Fig-14

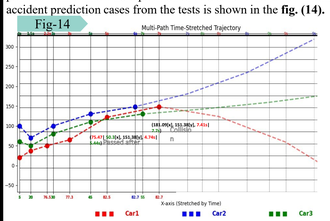
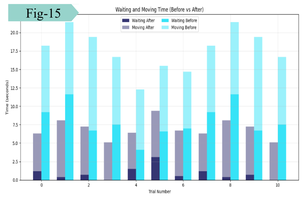


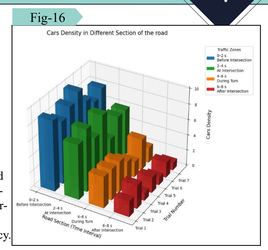
Fig-15



ANALYSIS

- Cars density: To analyize system efficiency in resolving congestion, cars density was calculated per each test. The following **Fig. (16)** shows cars density (no of cars/ section of area) for each trial divided into four stages: before intersection, at intersection, moving turn/ forward and after intersection. highest values were at intersections. However, due to using system on road in trials 2, 3 and 5, a quite decrease in cars density was detected proving system efficiency.

Fig-16



Negative results:

- Camera Calibration:
Although the camera's measurement error was small, it still caused noticeable shifts from expected values. This was improved by adding road reference checks and using IR sensors as a trusted comparison point. Regularly matching the AI's readings with sensor data and training the model on more examples helped boost accuracy.

- Camera Limitations:
When we tested the system in real-life conditions, the camera sometimes struggled—especially in rain, low light, or when dust settled on the lens. Reflections and unclear visuals occasionally caused incorrect detections, showing that the camera has limits in tougher environments.

- Latency Spikes Under High Load:
The system generally responded quickly, but during heavy congestion, response times sometimes dropped. These brief delays appeared only in the most crowded moments, showing how the system reaches its limits under very high pressure.

Learning Transfer

(MA.3.02): Regression helped us model the traffic data, allowing the regression curve to show the trend of car movement and calculate traffic flow more accurately.

(MA.3.07): The camera extracts a car's position, then derives velocity and acceleration to forecast movement, braking needs, and potential collisions.

(PH.3.04) : Communication system methods taught us how to send data reliably, which we used to ensure fast and accurate information transfer in our traffic system.

(CH.3.02) : Understanding uncertainty and error helped us choose properly calibrated tools like the vernier caliper to ensure accurate, reliable measurements for safe and effective analysis

(CS.3.01) : Helped create the mobile dashboard app that shows live sensor data, visualizes traffic, and provides an AI route assistant while also sending and receiving real-time data with the main system.

(ES.3.02) : The project's local positioning system was inspired by the GPS & Helped us determine precise positions, distance, and velocity

CONCLUSION

Through comprehensive testing and comparison with prior solutions, the intelligent traffic system designed in this study confirmed significant and measurable improvements in urban traffic quality and safety. While existing prior solutions such as the Cairo ITS and the New Administrative Capital ITS offer valuable monitoring capabilities and centralized control, research highlighted determined limitations in their high operational cost, restricted scalability, fixed-timing dependency, and lack of predictive accident detection or V2V communication. Addressing these gaps, the system performed well in all the design criteria. The change in Velocity-time and Position-time graphs in trip time of 58.3%, reduction in the wait time of 85.9% at the intersection points, and enhancement in traffic flow of ~300% supported by the accuracy of the system of ~90% in accurately measuring car density from the camera at 89.2% accuracy. The graphical trend of the performance of the velocity-time and distance-time clearly identified the adaptation of the traffic in reducing the number of stops in the traffic. Additionally, the accuracy of the prediction of accidents of 80% was obtained from the machine learning decisions. In addition, the system demonstrated effective treatment of speed-limit violations with detection accuracy of ~97%. The processing time response averaged approximately 1.46 seconds. Simulated evaluations further indicated decreases in fuel emission and pollution by 51.6%, supporting environmental objectives. Compared with prior solutions that require large setup, this automated design proved scalable and low-cost due to IoT communication and integrated components. Ultimately, the results confirm that an ICT-integrated, data-driven, and adaptively controlled road system can clearly enhance safety, trip time, vehicle movement continuity, emergency priority, and environmental outcomes, positioning the model as a practical and forward-looking contribution to addressing Egypt's traffic-congestion challenges and advancing transportation sustainability.

RECOMMENDATION

To further improve the efficiency, scalability, and reliability of the intelligent traffic system, the following modifications are suggested:

Infrastructure-Level Improvements:

I. Enhance Camera Robustness and Quality for Real-World Conditions:
Upgrading to weather-resistant, high-resolution cameras with improved low-light performance enhances detection reliability in rain, dust, night conditions, and reflective environments. Using lens-protection housings and anti-fog coatings significantly increases long-term stability and reduces environmental interference.

II. Install Smart Environmental Sensors to Adjust Traffic Decisions Based on Air Quality:
Integrating CO_2, NO_2, and particulate-matter sensors into intersections allows the system to adapt traffic-light timing when pollution levels increase. This supports better traffic dispersion, reduces emissions in c ongested areas, and enhances environmental monitoring capabilities.

Technology & AI Enhancements:

I. Upload the Server to a Secure Google Cloud Platform for Faster Processing:
Migrating decision-processing servers to a scalable cloud environment improves computing power, reduces latency, and ensures real-time responsiveness. Cloud-based infrastructure enhances security, availability, and reliability for large-scale, city-wide deployments.

II. Expand Machine Learning Training for Enhanced Vehicle Detection and Tracking:
Training the ML model on larger and more diverse datasets—including different vehicle types, lighting conditions, and congestion levels—significantly increases detection accuracy, tracking stability, and overall system intelligence, especially in complex intersections.

III. Improve the Fastest Routing Detection Algorithm by Including More Real-Time Factors:
Refining the routing engine to consider additional variables such as congestion density, predicted flow, emergency vehicle paths, and environmental conditions increases the accuracy of recommended routes and enhances overall traffic optimization.

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